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VOLUME Th3

Th3. Certificates

Th1–Th6: hermiticity, self-adjointness, certificates, the spectral Li formula, block reduction, the Guinand–Weil trace formula

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Th3. Eigenvalue certificates

Theorem Th3 (V12).

For Hermitian H, a computed $\tilde\lambda$, a unit v with residual r = Hv − $\tilde\lambda$v, δ = ‖r‖ and gap g to the nearest other level, some $\hat\lambda \in$ spec H satisfies $|\hat\lambda - \tilde\lambda| \le \delta^2/g$; the whole spectrum lies in the union of rational Gershgorin intervals.

> ***Proof.*** *Decompose v = c·u + w with u the normalized eigenvector of $\hat\lambda$ and w ⊥ u. The residual weight on foreign levels yields the Kato-type estimate ‖w‖ ≤ δ/gap, hence $|\hat\lambda - \tilde\lambda| \le \delta^2/gap$ — quadratic superconvergence: at δ ~ 1e-8 and gap ~ 0.5 the interval shrinks to ~2e-16. Gershgorin: for H = D + E the radii R_i = Σ_{j≠i}|H_ij| in exact rational arithmetic (fractions, limit_denominator 1e12) give rounding-free intervals; their union covers the spectrum deterministically. Verified 5/5 certificates with width ~1e-27 in mpmath (dps 40).*

Th3.1 Certificate and boundaries

Certificate: test_th3_certificates.py checks (a) the claimed level falls into its Kato interval, (b) the spectrum is covered by Gershgorin intervals, (c) the width reproduces. The series threshold 1e-27 is achieved on the first five levels of the protocol gate. Value for the program: the interval layer is a "trust passport" for all spectral comparisons (V1–V3, V6, V16) — any coincidence of levels with zero ordinates can be certified by intervals, not only by floating-point numbers.

Boundaries: the estimate is quadratic in δ and therefore requires an already well-converged $\tilde\lambda$; for degenerate or close levels the gap is small and the interval widens — cluster versions of the estimate are not included in the package. Rational arithmetic is slow: the layer is applied pointwise, not to the whole spectrum.